

Program 2

Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Prim's algorithm.

```
#include<stdio.h>
int n,c[20][20],i,j,visited[20];
void prim();

void main()
{
    printf("enter the no of vertices\n");
    scanf("%d",&n);
    printf("enter the cost matrix\n");
    for(i=1;i<=n;i++)
    {
        for(j=1;j<=n;j++)
            scanf("%d",&c[i][j]);
        visited[i]=0;
    }
    prim();
}

void prim()
{
    int min,b,a,count=0,cost=0;
    min=999;
    visited[1]=1;
    while(count<n-1)
    {
        min=999;
        for(i=1;i<=n;i++)
            for(j=1;j<=n;j++)
                if(visited[i]&&!visited[j]&&min>c[i][j])
                {
                    min=c[i][j];
                    a=i;
                    b=j;
                }
        printf("%d->%d=%d\n",a,b,c[a][b]);
        cost+=c[a][b];
        visited[b]=1;
        count++;
    }
    printf("\n total cost of spanning tree is %d",cost);
}
```

Sample Input and Output:

```
enter the no of vertices
4
enter the cost matrix
```

999 1 5 2
1 999 999 999
5 999 999 3
2 999 3 999
1->2=1
1->4=2
4->3=3

total cost of spanning tree is 6